Design a Hotel Management System

Let's design a hotel management system.

A Hotel Management System is a software built to handle all online hotel activities easily and safely. This System will give the hotel management power and flexibility to manage the entire system from a single online portal. The system allows the manager to keep track of all the available rooms in the system as well as to book rooms and generate bills.

BOOK ONLINE

System Requirements

We'll focus on the following set of requirements while designing the Hotel Management System:

1. The system should support the booking of different room types like standard, deluxe, family suite, etc.
2. Guests should be able to search the room inventory and book any available room.
3. The system should be able to retrieve information, such as who booked a particular room, or what rooms were booked by a specific customer.
4. The system should allow customers to cancel their booking - and provide them with a full refund if the cancelation occurs before 24 hours of the check-in date.
5. The system should be able to send notifications whenever the booking is nearing the check-in or check-out date.
6. The system should maintain a room housekeeping log to keep track of all housekeeping tasks.
7. Any customer should be able to add room services and food items.
8. Customers can ask for different amenities.
9. The customers should be able to pay their bills through credit card, check or cash.

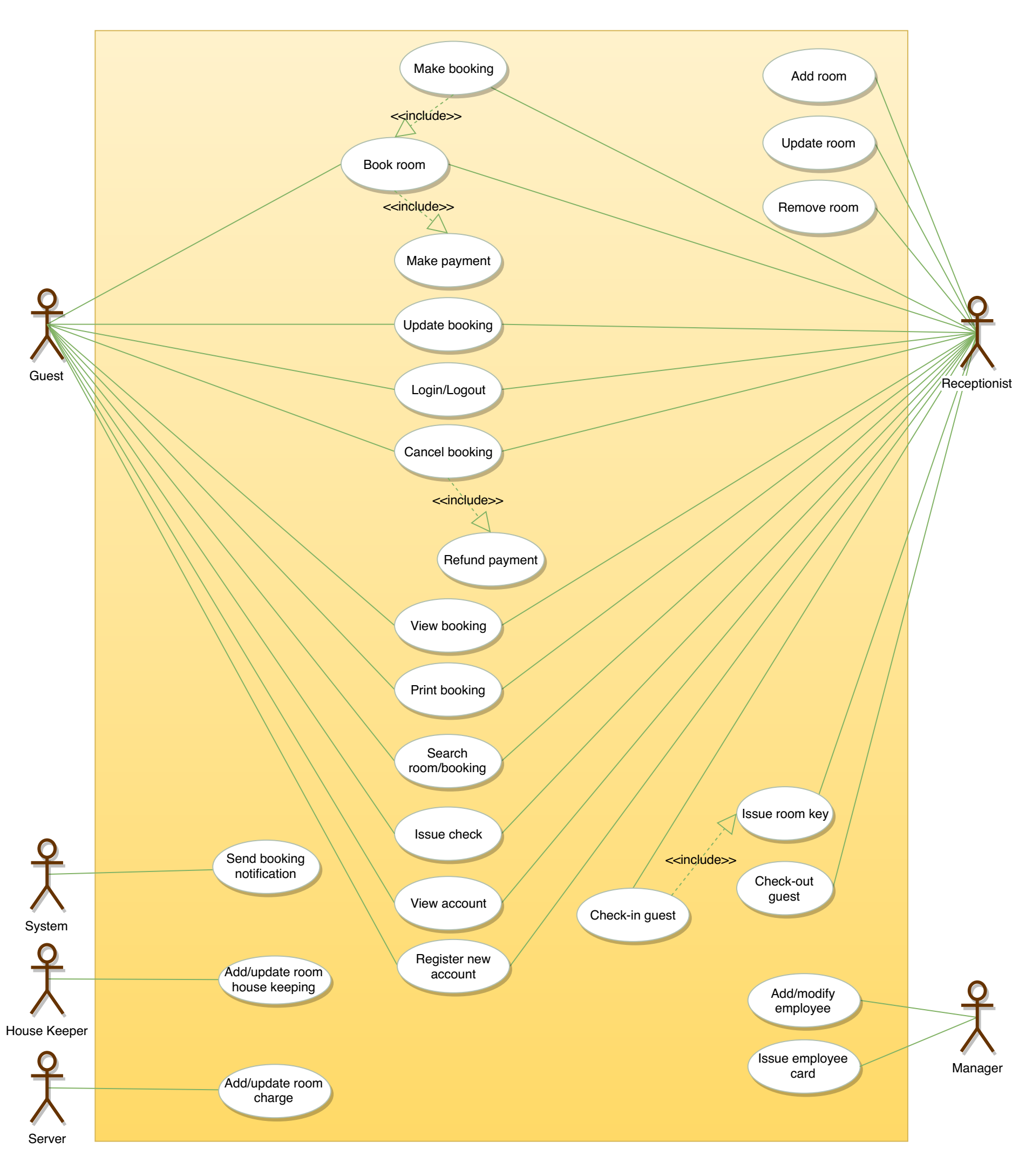
Use case diagram

Here are the main Actors in our system:

- **Guest**: All guests can search the available rooms, as well as make a booking.
- **Receptionist**: Mainly responsible for adding and modifying rooms, creating room bookings, check-in, and check-out customers.
- **System**: Mainly responsible for sending notifications for room booking, cancellation, etc.
- **Manager**: Mainly responsible for adding new workers.
- **Housekeeper**: To add/modify housekeeping record of rooms.
- **Server**: To add/modify room service record of rooms.

Here are the top use cases of the Hotel Management System:

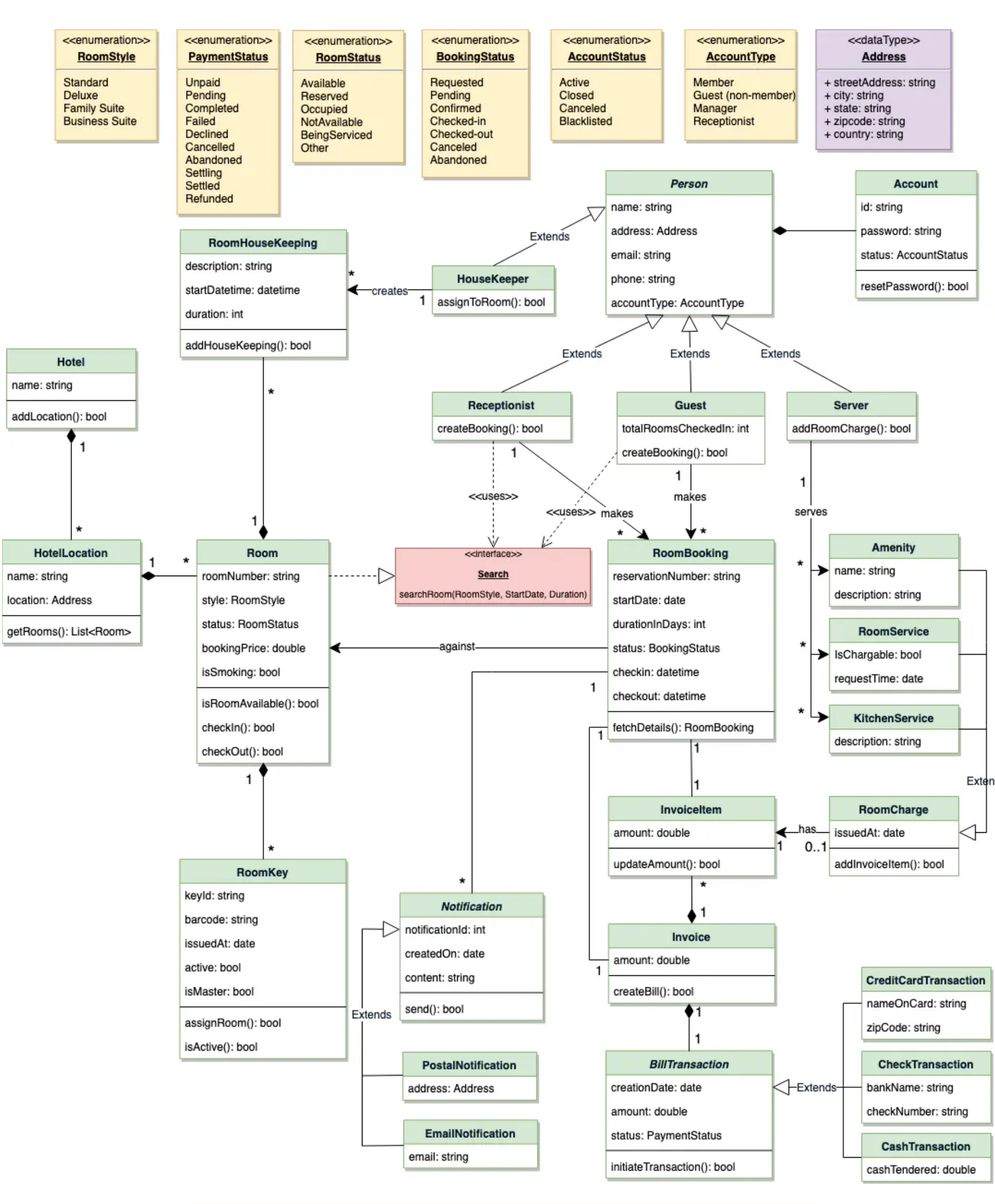
- **Add/Remove/Edit room**: To add, remove, or modify a room in the system.
- **Search room**: To search for rooms by type and availability.
- **Register or cancel an account**: To add a new member or cancel the membership of an existing member.
- **Book room**: To book a room.
- **Check-in**: To let the guest check-in for their booking.
- **Check-out**: To track the end of the booking and the return of the room keys.
- **Add room charge**: To add a room service charge to the customer's bill.
- **Update housekeeping log**: To add or update the housekeeping entry of a room.

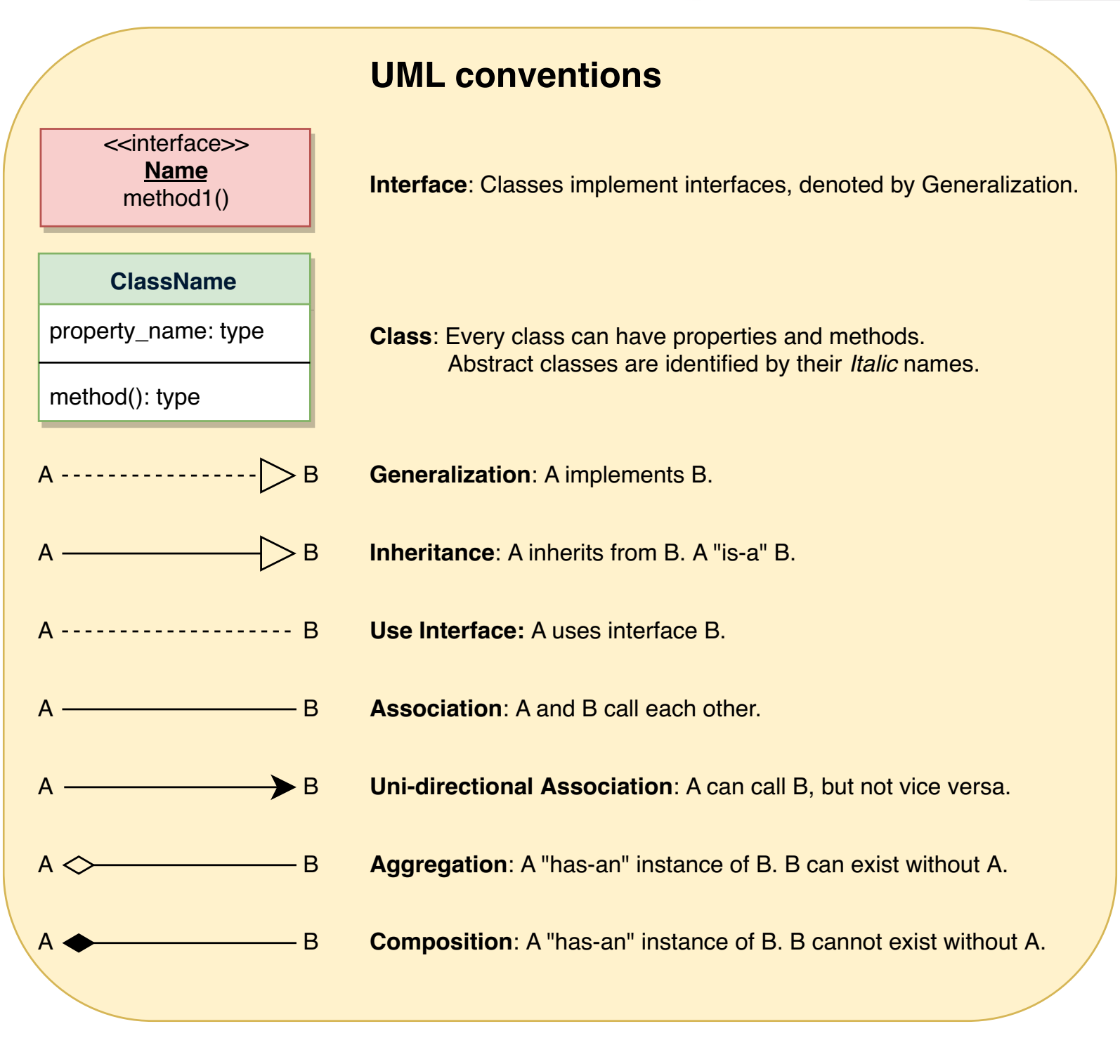

Make booking
<<include>>
Book room
<<include>>
Make payment
Update booking
Login/Logout
Cancel booking
<<include>>
Refund payment
View booking
Print booking
Search room/booking
Issue check
View account
Register new account
Add room
Update room
Remove room
Issue room key
<<include>>
Check-in guest
Check-out guest
Send booking notification
Add/update room house keeping
Add/update room charge
Add/modify employee
Issue employee card
Guest
Receptionist
System
House Keeper
Server
Manager

Class diagram

Here are the main classes of our Hotel Management System:

- **Hotel and HotelLocation**: Our system will support multiple locations of a hotel.
- **Room**: The basic building block of the system. Every room will be uniquely identified by the room number. Each Room will have attributes like Room Style, Booking Price, etc.
- **Account**: We will have different types of accounts in the system: one will be a guest to search and book rooms, another will be a receptionist. Housekeeping will keep track of the housekeeping records of a room, and a Server will handle room service.
- **RoomBooking**: This class will be responsible for managing bookings for a room.
- **Notification**: Will take care of sending notifications to guests.
- **RoomHouseKeeping**: To keep track of all housekeeping records for rooms.
- **RoomCharge**: Encapsulates the details about different types of room services that guests have requested.
- **Invoice**: Contains different invoice-items for every charge against the room.
- **RoomKey**: Each room can be assigned an electronic key card. Keys will have a barcode and will be uniquely identified by a key-ID.


<<enumeration>> RoomStyle
Standard
Deluxe
Family Suite
Business Suite
<<enumeration>> PaymentStatus
Unpaid
Pending
Completed
Failed
Declined
Cancelled
Abandoned
Settling
Settled
Refunded
<<enumeration>> RoomStatus
Available
Reserved
Occupied
NotAvailable
BeingServiced
Other
<<enumeration>> BookingStatus
Requested
Pending
Confirmed
Checked-in
Checked-out
Canceled
Abandoned
<<enumeration>> AccountStatus
Active
Closed
Canceled
Blacklisted
<<enumeration>> AccountType
Member
Guest (non-member)
Manager
Receptionist
<<dataType>> Address
+ streetAddress: string
+ city: string
+ state: string
+ zipcode: string
+ country: string
Person
name: string
address: Address
email: string
phone: string
accountType: AccountType
Account
id: string
password: string
status: AccountStatus
resetPassword(): bool
RoomHouseKeeping
description: string
startDatetime: datetime
duration: int
addHouseKeeping(): bool
HouseKeeper
assignToRoom(): bool
creates
Extends
Hotel
name: string
addLocation(): bool
Receptionist
createBooking(): bool
Guest
totalRoomsCheckedIn: int
createBooking(): bool
Server
addRoomCharge(): bool
<<uses>>
makes
serves
HotelLocation
name: string
location: Address
getRooms(): List<Room>
Room
roomNumber: string
style: RoomStyle
status: RoomStatus
bookingPrice: double
isSmoking: bool
isRoomAvailable(): bool
checkIn(): bool
checkOut(): bool
<<interface>> Search
searchRoom(RoomStyle, StartDate, Duration)
RoomBooking
reservationNumber: string
startDate: date
durationInDays: int
status: BookingStatus
checkin: datetime
checkout: datetime
fetchDetails(): RoomBooking
against
Amenity
name: string
description: string
RoomService
isChargable: bool
requestTime: date
KitchenService
description: string
RoomKey
keyId: string
barcode: string
issuedAt: date
active: bool
isMaster: bool
assignRoom(): bool
isActive(): bool
Notification
notificationId: int
createdOn: date
content: string
send(): bool
PostalNotification
address: Address
EmailNotification
email: string
InvoiceItem
amount: double
updateAmount(): bool
RoomCharge
issuedAt: date
addInvoiceItem(): bool
has
0..1
Invoice
amount: double
createBill(): bool
BillTransaction
creationDate: date
amount: double
status: PaymentStatus
initiateTransaction(): bool
CreditCardTransaction
nameOnCard: string
zipCode: string
CheckTransaction
bankName: string
checkNumber: string
CashTransaction
cashTendered: double


UML conventions
<<interface>>
Name
method1()
ClassName
property_name: type
method(): type
A
B

Interface: Classes implement interfaces, denoted by Generalization.

Class: Every class can have properties and methods. Abstract classes are identified by their *Italic* names.

Generalization: A implements B.

Inheritance: A inherits from B. A "is-a" B.

Use Interface: A uses interface B.

Association: A and B call each other.

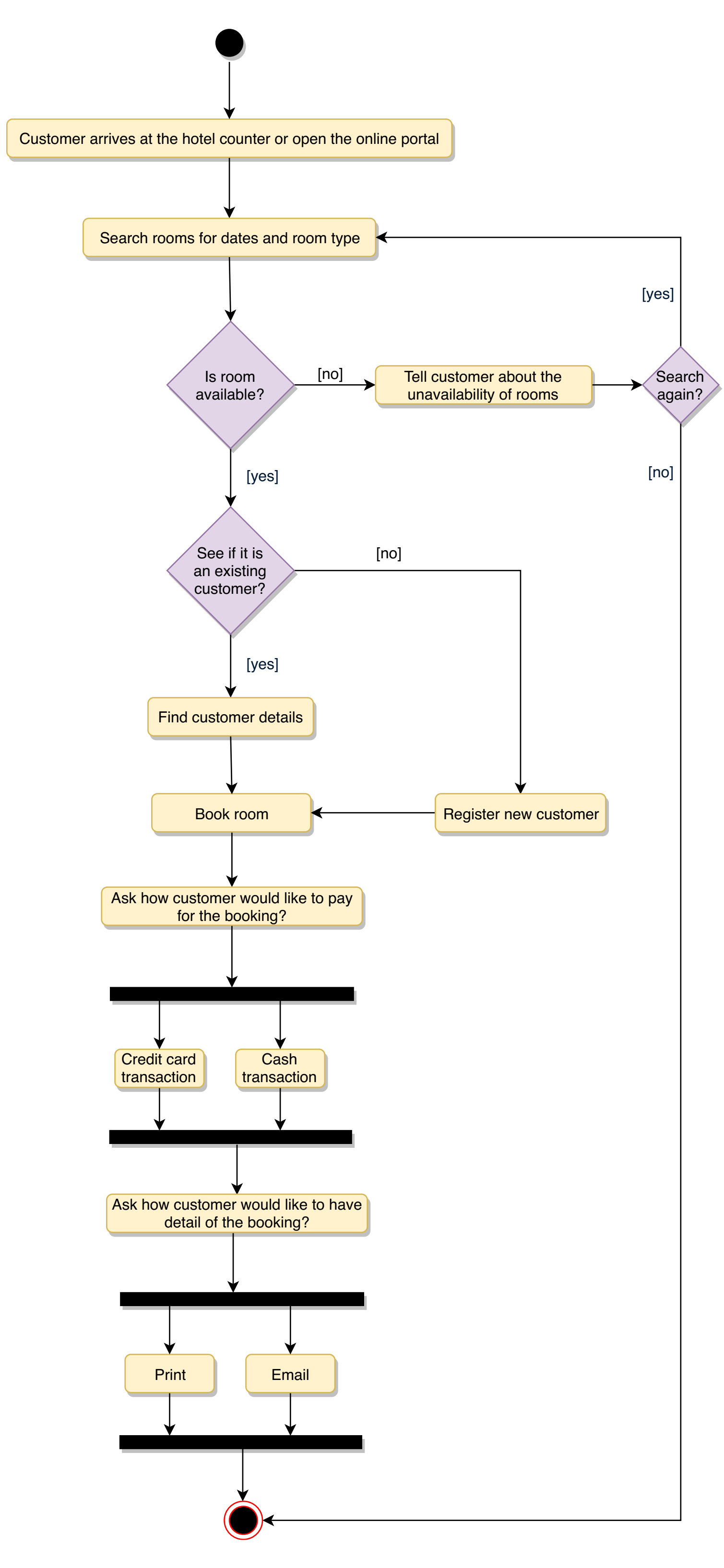
Uni-directional Association: A can call B, but not vice versa.

Aggregation: A "has-an" instance of B. B can exist without A.

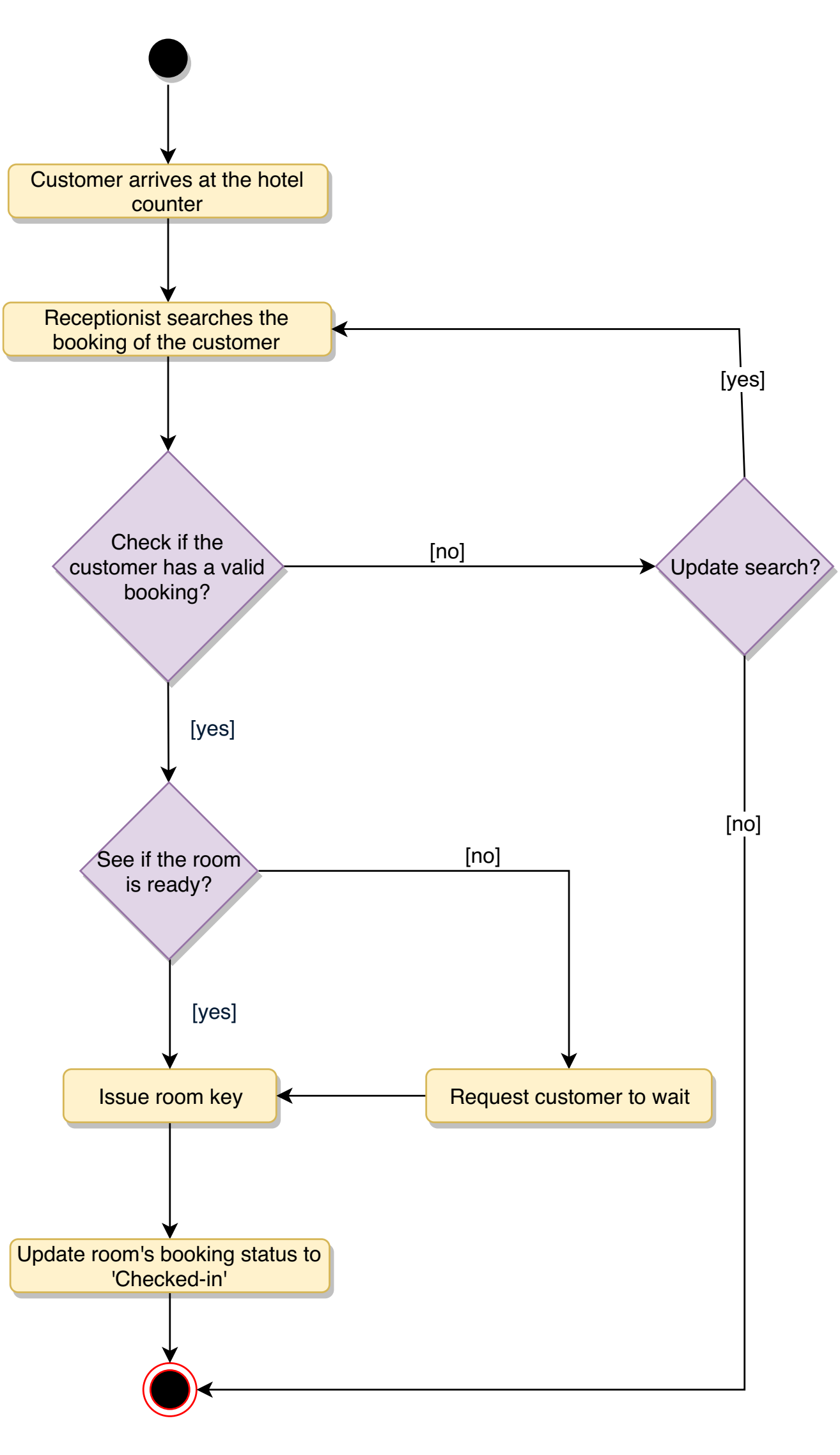
Composition: A "has-an" instance of B. B cannot exist without A.

Activity diagrams

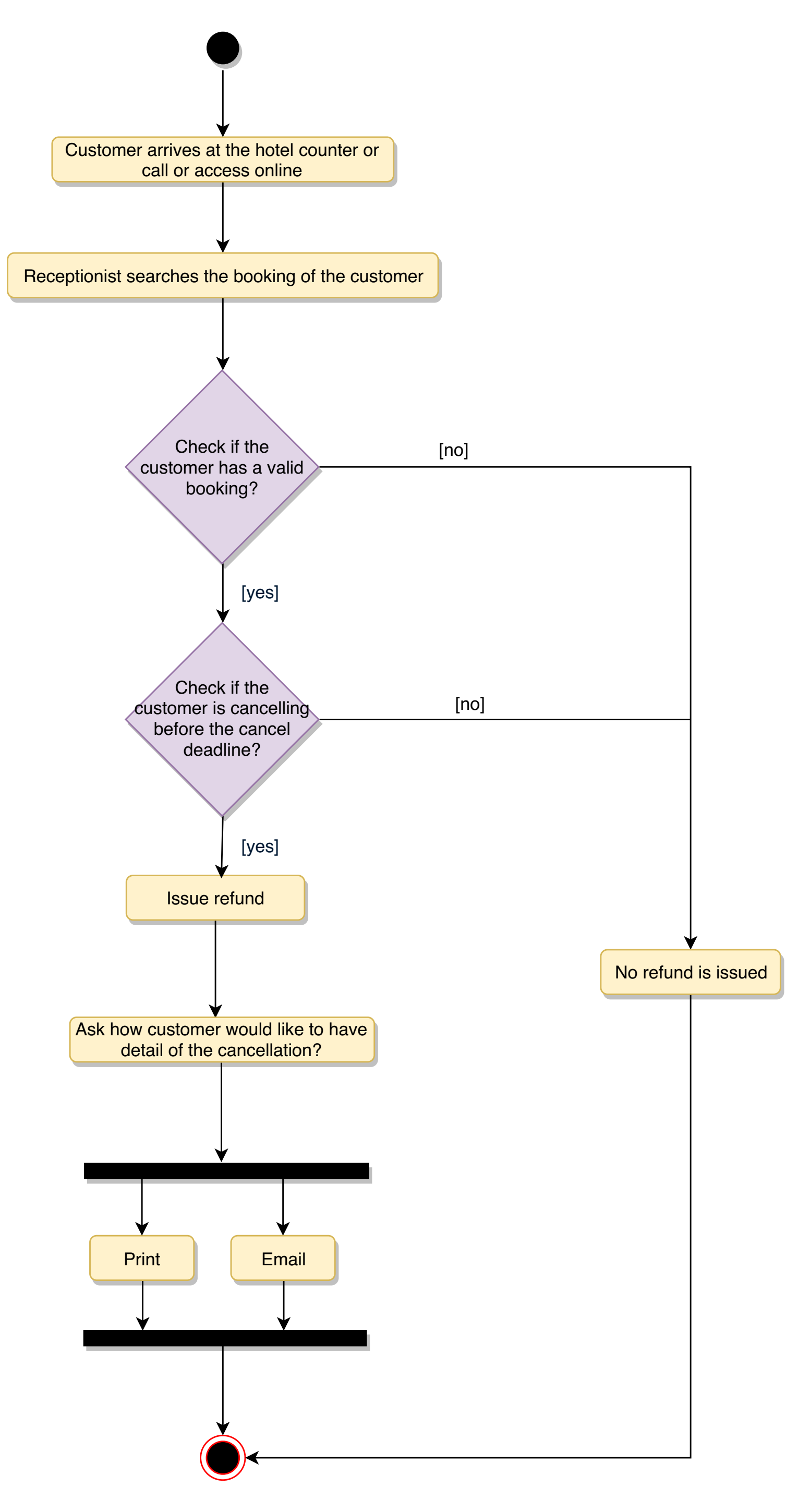
Make a room booking: Any guest or receptionist can perform this activity. Here are the set of steps to book a room:


Customer arrives at the hotel counter or open the online portal
Search rooms for dates and room type
Is room available?
[no]
Tell customer about the unavailability of rooms
Search again?
[yes]
[no]
[yes]
See if it is an existing customer?
[no]
[yes]
Find customer details
Book room
Register new customer
Ask how customer would like to pay for the booking?
Credit card transaction
Cash transaction
Ask how customer would like to have detail of the booking?
Print
Email

Check in: Guest will check in for their booking. The Receptionist can also perform this activity. Here are the steps:


Customer arrives at the hotel counter
Receptionist searches the booking of the customer
[yes]
Check if the customer has a valid booking?
[no]
Update search?
[yes]
[no]
See if the room is ready?
[no]
[yes]
Issue room key
Request customer to wait
Update room's booking status to 'Checked-in'

Cancel a booking: Guest can cancel their booking. Receptionist can perform this activity. Here are the different steps of this activity:


Customer arrives at the hotel counter or call or access online
Receptionist searches the booking of the customer
Check if the customer has a valid booking?
[no]
[yes]
Check if the customer is cancelling before the cancel deadline?
[no]
[yes]
Issue refund
No refund is issued
Ask how customer would like to have detail of the cancellation?
Print
Email

Code

Here is the high-level definition for the classes described above.

Enums, data types, and constants: Here are the required enums, data types, and constants:

```java
public enum RoomStyle {
    STANDARD, DELUXE, FAMILY_SUITE, BUSINESS_SUITE
  }

  public enum RoomStatus {
    AVAILABLE, RESERVED, OCCUPIED, NOT_AVAILABLE, BEING_SERVICED, OTHER
  }

  public enum BookingStatus {
    REQUESTED, PENDING, CONFIRMED, CHECKED_IN, CHECKED_OUT, CANCELLED, ABANDONED
  }

  public enum AccountStatus {
    ACTIVE, CLOSED, CANCELED, BLACKLISTED, BLOCKED
  }

  public enum AccountType {
    MEMBER, GUEST, MANAGER, RECEPTIONIST
  }

  public enum PaymentStatus {
```

Account, Person, Guest, Receptionist, and Server: These classes represent the different people that interact with our system:

```java
// For simplicity, we are not defining getter and setter functions. The reader can
    // assume that all class attributes are private and accessed through their respectiv
    // public getter method and modified only through their public setter method.

    public class Account {
      private String id;
      private String password;
      private AccountStatus status;

      public boolean resetPassword();
    }

    public abstract class Person {
      private String name;
      private Address address;
      private String email;
      private String phone;

      private Account account;
    }

```

Hotel and HotelLocation: These classes represent the top-level classes of the system:

```java
public class HotelLocation {
  private String name;
  private Address location;

  public List<Room> getRooms();
  }

  public class Hotel {
  private String name;
  private List<HotelLocation> locations;

  public boolean addLocation(HotelLocation location);
  }

```

Room, RoomKey, and RoomHouseKeeping: To encapsulate a room, room key, and housekeeping:

```java
public interface Search {
    public static List<Room> search(RoomStyle style, Date startDate, int duration);
  }

  public class Room implements Search {
    private String roomNumber;
    private RoomStyle style;
    private RoomStatus status;
    private double bookingPrice;
    private boolean isSmoking;

    private List<RoomKey> keys;
    private List<RoomHouseKeeping> houseKeepingLog;

    public boolean isRoomAvailable();
    public boolean checkIn();
    public boolean checkOut();

    public static List<Room> search(RoomStyle style, Date startDate, int duration) {
      // return all rooms with the given style and availability
    }
```

RoomBooking and RoomCharge: To encapsulate a booking and different charges against a booking:

```java
public class RoomBooking {
  private String reservationNumber;
  private Date startDate;
  private int durationInDays;
  private BookingStatus status;
  private Date checkin;
  private Date checkout;

  private int guestID;
  private Room room;
  private Invoice invoice;
  private List<Notification> notifications;

  public static RoomBooking fectchDetails(String reservationNumber);
}

public abstract class RoomCharge {
  public Date issueAt;
  public boolean addInvoiceItem(Invoice invoice);
}

```

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